

Göksenin Yüksel

PhD Candidate in AI · Machine hearing, representation learning and spatial audio

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Research Interests

Machine hearing: Models that understand real-world sound scenes: what is heard and where it comes from

Representation learning: Self-supervised learning (masked autoencoders, JEPA), audio foundation models, robustness to noise and reverberation

Spatial audio: Ambisonics, binaural audio, geometry-agnostic spatial representations for arbitrary microphone arrays, sound event localization and detection (SELD), equivariant models, spatial audio large language models

Speech: Compute-efficient self-supervised speech representation learning, automatic speech recognition (ASR)

Education

PhD	Donders Institute, Radboud University , Artificial Intelligence, Machine Hearing • Expected graduation: January 2028. Research: machine hearing, self-supervised (spatial) audio and speech representation learning, and spatial audio LLMs. • AudioSphere (NeurIPS 2026): first sound event localization and detection from a frozen self-supervised backbone; 1.9° median direction-of-arrival error on a planar microphone array. • SpeechJEPA beats wav2vec 2.0 with one hour of labels while using about 17× fewer training FLOPs. • Bearings raises the location-aware SELD F-score from below 4 to 50 on TAU-NIGENS 2021. • Supervisors: Kiki van der Heijden and Marcel van Gerven. Member of the ELLIS Unit Nijmegen.	Nijmegen, NL Sept 2024 – Jan 2028
MSc	University of Amsterdam , Artificial Intelligence • GPA: 4.0/4.0. • Thesis: Domain Adaptation of Dense Retriever Models.	Amsterdam, NL Sept 2022 – Sept 2024
BSc	Radboud University , Artificial Intelligence • GPA: 3.96/4.0. • Teaching assistant for Functional Programming and Bayesian Statistics.	Nijmegen, NL Sept 2019 – Sept 2022

Grants

Snellius compute grants: Three grants (EINF-17264, EINF-14624, EINF-12218) totaling 3 million SBUs, about 15,000 H100 GPU-hours on the Dutch national supercomputer (SURF)

Peer-Reviewed Publications

AudioSphere: Towards Self-Supervised Spatial Audio Representation Models Göksenin Yüksel, Marcel van Gerven, Kiki van der Heijden github.com/labhamlet/AudioSphere (NeurIPS 2026)	Sept 2026
University of Amsterdam at the CLEF 2024 SimpleText Track Jan Bakker, Göksenin Yüksel, Jaap Kamps ceur-ws.org/Vol-3740/paper-310.pdf (Working Notes of CLEF 2024, CEUR Workshop Proceedings 3740)	Sept 2024
Style Randomization Improves the Robustness of Breast Density Estimation in MR Images Göksenin Yüksel, Koen Eppenhof, Jaap Kroes, Marcel Worring proceedings.mlr.press/v250/yuksel24a.html (MIDL 2024 (PMLR 250))	July 2024
Reproducibility Study of “Quantifying Societal Bias Amplification in Image Captioning” Farrukh Barotov*, Göksenin Yüksel*, Darie Petcu*, Jan Bakker* openreview.net/forum?id=eJmQJT0Dtt (ML Reproducibility Challenge 2022, poster at NeurIPS 2023; * equal contribution)	Dec 2023

Under Review

SpeechJEPA: Clustering-Free Latent Prediction for Compute-Efficient Speech Representation Learning Göksenin Yüksel , Marcel van Gerven, Kiki van der Heijden github.com/labhamlet/SpeechJEPA (Under review at ICASSP 2027)	Sept 2026
Bearings: Self-Supervised Soundfield Embeddings from First-Order Ambisonics Göksenin Yüksel , Marcel van Gerven, Kiki van der Heijden arxiv.org/abs/2609.23152 (Under review at ICASSP 2027)	Sept 2026
EquiSELD: Efficient Training of Equivariant Sound Event Localization and Detection Networks Göksenin Yüksel , Marcel van Gerven, Kiki van der Heijden arxiv.org/abs/2609.23156 (Under review at ICASSP 2027)	Sept 2026
WavJEPA: Semantic Learning Unlocks Robust Audio Foundation Models for Raw Waveforms Göksenin Yüksel , Pierre Guetschel, Michael Tangermann, Marcel van Gerven, Kiki van der Heijden arxiv.org/abs/2509.23238 (Under review at IEEE/ACM TASLP)	Sept 2025

Preprints

GRAM: Spatial General-Purpose Audio Representations Göksenin Yüksel , Marcel van Gerven, Kiki van der Heijden arxiv.org/abs/2506.00934	June 2025
Remining Hard Negatives for Generative Pseudo Labeled Domain Adaptation Göksenin Yüksel , David Rau, Jaap Kamps arxiv.org/abs/2501.14434	Jan 2025
Interpretability Analysis of Domain Adapted Dense Retrievers Göksenin Yüksel , Jaap Kamps arxiv.org/abs/2501.14459	Jan 2025
On Correlating Factors for Domain Adaptation Performance Göksenin Yüksel , Jaap Kamps arxiv.org/abs/2501.14466	Jan 2025

Experience

ScreenPoint Medical , AI Research Intern - Built a robust 3D MRI segmentation pipeline for breast density estimation, now used for internal R&D. - Proposed a style-randomization augmentation for domain generalization across MRI sequences; first-author paper at MIDL 2024.	Nijmegen, NL Sept 2023 – Feb 2024 6 months
ScreenPoint Medical , Software Engineer (part-time) - Developed and maintained automated tests for large-scale medical AI software (2D/3D medical image processing). - Automated manual verification steps and led a project to produce a shorter, more maintainable verification report.	Nijmegen, NL Jan 2023 – Sept 2024 1 year 9 months
Arcadis , Data Science Intern - Built and deployed machine learning dashboards for housing price analysis and for the optimal placement of EV parking spots and charging poles. - Presented findings to stakeholders and advised on data-driven business decisions (Python, SQL).	Netherlands July 2021 – Oct 2021 4 months

Technical Skills

Deep learning: PyTorch, PyTorch Lightning, Transformers, FlashAttention, large language models (LLMs)
Distributed training: Large-scale GPU cluster training; efficient multi-node scaling with FSDP and DDP via torchrun
Audio & signal processing: Ambisonics, binaural audio, microphone arrays, room acoustics simulation
Languages & tools: Python, Java, Bash, Linux, Docker, Git, Apache Spark; AI coding agents (Claude Code, Codex)